FLEXICLEAN: A MOBILE-CONTROLLED SMART CLEANING ROBOT FOR EFFICIENT AND COST-EFFECTIVE AUTOMATION

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***Abstract*— Maintaining cleanliness in homes and small businesses is essential but often requires significant manual effort, especially for cleaning both floors and vertical surfaces like walls. Traditional cleaning methods are time-consuming, labor-intensive, and inefficient in reaching corners or elevated areas, while most commercial robotic cleaners are limited to floor cleaning and consume excessive resources. This project, FlexiClean, aims to design and develop a dual-surface smart cleaning robot capable of cleaning both floors and walls, controlled remotely via a mobile application.**

***Keywords*— Smart Cleaning Robot, Dual-Surface Cleaning, Floor and Wall Cleaning, Mobile App Control, Autonomous Navigation, Ultrasonic Sensors, IoT-Based Cleaning System, Robotics Automation**

I. INTRODUCTION

In the modern era of technological advancement and rapid urbanization, automation has become a key enabler in enhancing comfort, productivity, and hygiene. Among various applications of automation, cleaning and maintenance represent one of the most essential yet time-consuming household and commercial tasks. The increasing demand for efficient, hygienic, and effortless cleaning solutions has accelerated the development of robotic cleaning systems. However, most commercially available robotic cleaners are designed primarily for horizontal floor cleaning and are incapable of cleaning vertical surfaces such as walls, glass panels, and ceilings. This limitation leaves a significant portion of interior environments uncleaned and requires manual intervention, which defeats the goal of full automation.
Traditional cleaning approaches involve physical effort, use of excess water and detergents, and limited access to elevated or cornered areas. Moreover, high-end robotic systems that attempt to perform both wall and floor cleaning are often expensive and unsuitable for small-scale domestic or business use. These gaps highlight the need for an affordable, energy-efficient, and versatile cleaning system capable of operating across multiple surfaces without complex setup or supervision.

FlexiClean is designed to address these limitations by introducing a mobile-controlled smart cleaning robot that performs dual-surface cleaning — effectively handling both floor and wall cleaning operations through an intelligent control mechanism. The robot integrates ultrasonic sensors for real-time obstacle detection and Bluetooth-based control via a dedicated mobile application, allowing users to manage movement, cleaning modes, and spray functions wirelessly. The system employs an optimized cleaning mechanism combining rotating brushes, liquid spraying, and suction, ensuring thorough and uniform cleaning with minimal water and detergent usage.

The mechanical structure of FlexiClean is engineered for dual orientation, enabling smooth transitions between horizontal and vertical cleaning. The use of a microcontroller-based architecture (Arduino Nano) ensures precise control of motors, sensors, and pumps, while a Li-ion power source provides efficient energy management. The robot's modular design allows easy maintenance and potential scalability for additional functionalities such as voice commands, AI-based dirt detection, and IoT integration.

The primary objectives of this project are to design a cost-effective, eco-friendly, and user-friendly cleaning robot that minimizes manual labor, enhances cleaning efficiency, and supports sustainable living. The system aims to bridge the gap between affordability and functionality by offering smart navigation, flexible operation, and ease of use. Furthermore, the integration of mobile connectivity brings automation closer to users' fingertips, aligning with the growing trend of smart home ecosystems.

By combining mechanical innovation, sensor integration, and intelligent control, FlexiClean contributes to the advancement of home automation technology and demonstrates the potential of low-cost robotics in addressing real-world cleanliness challenges in households, educational institutions, and small businesses.

II. LITERATURE REVIEW

Researchers have progressively advanced cleaning robots from simple autonomous floor units to complex multi-surface systems with intelligent sensing, control, and connectivity. Muthugala *et al.* [1] investigated the design and control of wall-cleaning robots with adhesion awareness, highlighting challenges in maintaining contact force and stability on vertical surfaces. Their work laid foundational insights into adhesion mechanisms crucial for wall operation. Building on multi-surface adaptability, Rahman and Chowdhury [2] proposed a hybrid robot capable of both floor and wall cleaning. Their architecture allowed surface transition but lacked advanced autonomous navigation and environment awareness. Sensor-based autonomy has been a consistent theme in the literature. Several studies adopted ultrasonic sensors for obstacle detection and navigation in robotic cleaners. IJERT research [3] developed an Arduino-based autonomous floor cleaner with ultrasonic obstacle avoidance, focusing on low-cost hardware and basic path planning. Similarly, JETIR authors [4] presented a Bluetooth-controlled cleaning robot using ultrasonic sensors for obstacle avoidance and interactive control. Integration of cleaning mechanisms was further explored in other works. IJERT [12] and ResearchGate [10] described automated floor cleaners with integrated spray, brush, and pump systems, enabling simultaneous scrubbing and debris removal. These systems emphasized functional cleaning performance together with sensor-based navigation. Connectivity and user interfacing have been addressed through Bluetooth and IoT technologies. Hybrid hardware projects such as the IoT-enabled smart cleaner [9] demonstrated remote control and monitoring capabilities, enhancing user interaction beyond traditional autonomous operation. From a design methodology perspective, studies such as that by MDPI [6] applied Kano and Quality Function Deployment (QFD) tools to prioritize features and align robot functionality with user expectations. This approach illustrates how customer-centric design principles can inform robotic development beyond technical innovation.Surveys and analytical reviews have been conducted to organize the broad range of sensors and control strategies available for cleaning robots. Kumar and Singh [7] provided an overview of sensor types and intelligent control methods, underscoring the importance of effective perception in robot autonomy. SSRN research [11] specifically targeted enhancements in ultrasonic sensing accuracy, addressing key limitations in traditional sensing for robust obstacle detection. Further advancements include adaptive planning strategies. For instance, approaches leveraging human density surveillance for dynamic path planning [8] introduced context-aware navigation, which is particularly relevant in environments shared with humans. Complementary research on climbing robotics [5,14,15] informed the vertical mobility aspects of cleaning robots. This includes suction and grip analysis [5], modular plane transitions [14], and systematic categorization of adhesion methods (magnetic, vacuum, bio-inspired) relevant for wall-cleaning mechanisms [15]. Collectively, the literature demonstrates a clear trend toward multi-functional, sensor-rich, and connected cleaning robots. However, many existing systems are either limited to floor cleaning or lack advanced autonomy and efficient integration of multi-surface operation

III. PROPOSED SYSTEM

The proposed system, FlexiClean, is designed to provide an affordable, efficient, and dual-surface smart cleaning solution for homes and small commercial spaces. The methodology focuses on integrating mechanical design, sensor-based navigation, mobile control, and fluid management to achieve autonomous cleaning on both floors and walls with minimal human intervention.

A. System Overview

FlexiClean comprises three major subsystems: hardware, software, and control interface. The hardware includes a dual-orientation chassis, drive and cleaning motors, ultrasonic sensors, fluid dispensing pumps, and optional adhesion mechanisms such as suction fans or magnetic wheels. The software subsystem involves Arduino firmware, mobile application for Bluetooth control, and design/prototyping tools. The control interface provides the link between the user and the robot, allowing commands for movement, cleaning modes, and fluid spraying.

The core objective of the system is to enable autonomous navigation, dual-surface cleaning, and efficient resource usage, ensuring minimal water and detergent consumption. The methodology emphasizes modularity, allowing easy maintenance, component replacement, and potential upgrades.

B. Working Principle

The operation of the system begins with initialization and setup through the mobile application. Upon activation, the microcontroller initializes the sensors, motors, and pump, and verifies power levels, sensor calibration, and cleaning fluid availability before commencing operation. The user can then select between floor cleaning mode and wall cleaning mode using the mobile interface. In floor mode, the drive motors enable movement over horizontal surfaces while the cleaning actuator rotates the brush to scrub the floor. In wall mode, the adhesion mechanisms are activated and motor control is adjusted to support vertical movement while maintaining stability.

For navigation and obstacle avoidance, the system utilizes HC-SR04 ultrasonic sensors to detect obstacles in real time. The sensor data is processed by the Arduino Nano to determine distances from surrounding objects, and the motor driver accordingly adjusts the robot's direction by stopping or rerouting to prevent collisions. The cleaning mechanism consists of a DC motor-driven spinning brush for floor cleaning and rotating pads for wall surfaces. A 12 V diaphragm pump dispenses controlled amounts of water and detergent, synchronized with brush movement

to maximize cleaning efficiency while minimizing resource consumption.

The chassis design enables seamless transition between horizontal and vertical orientations. Magnetic wheels or suction-based adhesion systems provide stability during wall cleaning, while the microcontroller regulates motor speed and orientation to adapt to different surface inclinations. Throughout operation, the mobile application provides real-time feedback, including battery status, cleaning progress, and alerts for obstacles or low fluid levels. Users can pause, stop, or redirect the robot as required, with Bluetooth communication ensuring low-latency and reliable control.

C. System Flow

The operation flow of FlexiClean follows these steps: User input → Mode selection → Initialization → Obstacle detection → Motor and cleaning control → Fluid dispensing → Dual-surface adaptation → Completion → Return to standby. This sequential process ensures autonomous, intelligent cleaning across multiple surfaces with minimal human supervision.

D. Advantages of the Proposed System

The proposed system is designed to be cost-effective by utilizing low-cost components without compromising functionality. It supports dual-surface cleaning, enabling it to operate efficiently on floors, walls, and other vertical surfaces. The system is resource-efficient, with optimized water and detergent consumption to minimize waste. It is mobile-controlled, offering a user-friendly interface with real-time operational feedback. In addition, obstacle detection and adhesion mechanisms enhance safety and adaptability by ensuring stable performance across different surfaces. Furthermore, its modular and upgradable architecture allows both hardware and software enhancements, including the integration of advanced features such as AI-based dirt detection and IoT connectivity.

IV. SYSTEM ARCHITECTURE

The FlexiClean system architecture integrates hardware, software, and control interfaces to achieve efficient dual-surface cleaning with minimal human intervention. The architecture is designed to provide modularity, scalability, and reliability, allowing the robot to operate on both horizontal and vertical surfaces while maintaining stability, efficient navigation, and precise cleaning performance.

A. Hardware Architecture The hardware of FlexiClean enables efficient dual-surface cleaning with minimal human intervention. The system includes several modules, each chosen for specific functionality to ensure smooth operation on floors and walls.

1) Microcontroller: Arduino Nano (ATmega328P) Acts as the central processing unit, processing sensor inputs, controlling motors, and coordinating cleaning tasks. Provides multiple I/O pins for interfacing with sensors and actuators.



Fig. 1. Microcontroller: Arduino Nano (ATmega328P)

2) Connectivity Module: HC-05 Bluetooth Enables wireless control via a mobile application, supporting bidirectional communication for commands such as movement, cleaning mode selection, and fluid dispensing.

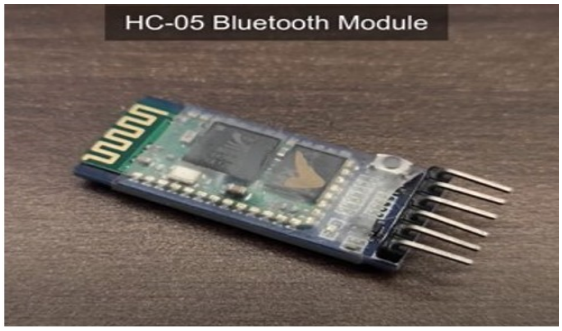


Fig. 2. Connectivity Module: HC-05 Bluetooth

3) Motor Driver: L293D / L298N Facilitates bidirectional DC motor control and protects the microcontroller from high current. Controls both drive and cleaning actuator motors.

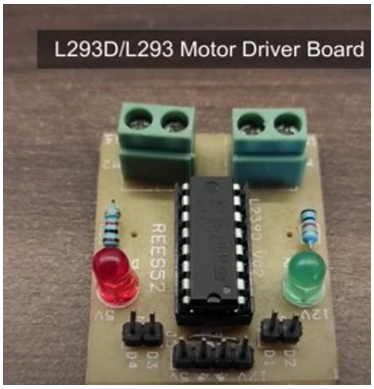


Fig. 3. Motor Driver: L293D / L298N

4) Drive Motors: 2 × 100 RPM DC Geared Motors Provide locomotion for floors and assist in vertical movement. Geared design ensures sufficient torque for different surfaces.



Fig. 4. Drive Motors

5) Sensors: 3 × HC-SR04 Ultrasonic Sensors
Enable obstacle detection and navigation, covering forward, side, and rear directions to prevent collisions.



Fig. 7. Sensors

6) Pump: 12 V Diaphragm Water Pump
Supplies controlled water and detergent, optimized for minimal usage. Can be synchronized with the cleaning brush for effective scrubbing.



Fig. 6 Pump

7) Cleaning Actuator: DC Motor-Powered Spinning Brush
Performs floor and wall cleaning. The rotating brush removes dirt efficiently and works with the fluid dispensing system.

8) Power Supply: 12 V Li-ion Battery Pack with BMS
Provides stable power for all components. The BMS ensures safe charging, discharging, and protection against overcurrent.

9) Chassis & Reservoirs: Dual-Orientation Frame with Tank
Supports floor and wall cleaning modes. Integrated water/soap tank feeds the pump, and the frame ensures stability and adhesion during vertical operation.

B. Software Architecture

The software of FlexiClean is responsible for integrating control, navigation, and user interface functionalities to enable efficient dual-surface cleaning. The system relies on a combination of firmware, mobile application, and design tools to ensure smooth operation and real-time responsiveness.

1) Firmware IDE: Arduino IDE (C/C++) :
The firmware is developed using the Arduino IDE, programmed in C/C++. It controls the microcontroller, processes sensor inputs, manages motor drivers, and coordinates the cleaning mechanism. Algorithms for obstacle detection, dual-surface switching, and fluid dispensing are embedded in the firmware to ensure precise and autonomous operation.

2) Mobile App: MIT App Inventor / Android Studio
A mobile application provides the user interface for remote control via Bluetooth. It enables commands for directional movement, cleaning mode selection, and activation of water/soap dispensing. The app ensures intuitive control, allowing minimal user intervention during operation.

3) Design Tools: Fritzing, Tinkercad, CAD
Software design tools such as Fritzing and Tinkercad are used for circuit simulation and prototyping, while CAD tools assist in designing the chassis and component layout. These tools ensure precise integration between hardware and software.

4) Version Control: Git/GitHub
Git is employed for version control, enabling systematic tracking of firmware updates and software modifications. Collaboration and iterative testing are facilitated through GitHub repositories.

5) Testing Tools: Serial Monitor, Bluetooth Terminal Apps
Serial Monitor and Bluetooth terminal applications are used for debugging and monitoring sensor data, motor responses, and app communication during development and testing.The combination of these software components allows FlexiClean to achieve reliable dual-surface cleaning with seamless user interaction and robust obstacle avoidance.

C. System Integration and Workflow

The system workflow follows a modular integration approach. Sensor data is continuously processed by the microcontroller, which adjusts motor commands and fluid dispensing accordingly. The mobile app provides a user-friendly interface for mode selection and control. Adhesion mechanisms and chassis design ensure dual-surface adaptability, while the power system maintains uninterrupted operation. This integrated architecture

allows autonomous navigation, efficient cleaning, and real-time user feedback, making FlexiClean a reliable solution for modern homes and small businesses.

V. MODULES AND EXPLANATION

The FlexiClean smart cleaning robot is designed as a modular system where each subsystem performs a specific role to ensure efficient and intelligent dual-surface cleaning. The following modules collectively enable seamless operation, mobility, and cleaning efficiency.

A. Mobile Control Module

The mobile control module enables users to remotely operate and monitor the robot using a Bluetooth-based Android application. Through the app, users can control movement directions—forward, backward, left, and right—along with activating or deactivating the cleaning mechanism and fluid dispenser. The Bluetooth communication, managed via the HC-05 module, ensures reliable short-range connectivity with minimal latency. The mobile interface also allows mode switching between floor and wall cleaning. This design eliminates the need for physical interaction, providing convenience and flexibility in operation.

B. Navigation and Obstacle Detection Module

The navigation and obstacle detection module employs three HC-SR04 ultrasonic sensors strategically positioned on the front and sides of the chassis. These sensors continuously emit and receive ultrasonic waves to detect objects in the path of the robot. The microcontroller processes the distance data and adjusts motor control commands to prevent collisions. This module plays a vital role in enabling autonomous navigation and smooth movement around obstacles. It enhances safety, prevents damage to surroundings, and allows efficient coverage of the cleaning area.

C. Cleaning Mechanism Module

The cleaning mechanism consists of a DC motor-powered rotating brush or mop designed for both horizontal and vertical surfaces. The rotating action effectively scrubs dust and stains from the surface. During floor cleaning, the brush ensures even contact with the ground, while during wall cleaning, specially designed pads maintain adhesion and consistent cleaning motion. The motor speed can be adjusted depending on the surface type. This module ensures effective cleaning with minimal manual intervention and is synchronized with the fluid dispensing system for coordinated operation.

D. Water and Detergent Dispensing Module

The fluid dispensing module uses a 12 V diaphragm pump connected to a nozzle system for controlled spraying of water and detergent. The quantity of liquid dispensed is optimized to minimize waste and maintain eco-friendly operation. The system can switch between water-only and detergent modes, depending on the cleaning requirement. The dispensing action is controlled by the microcontroller and synchronized with the brush's rotation, ensuring efficient removal of dirt while conserving resources. This module significantly contributes to the sustainability and performance of the FlexiClean system.

E. Motor Control Module

The motor control module utilizes the L293D or L298N motor driver to control both the drive motors and the cleaning actuator motors. These drivers handle the required current for the motors and provide bidirectional motion control, enabling forward and reverse movement. The drivers also protect the Arduino Nano from current surges, ensuring stable operation. Pulse Width Modulation (PWM) signals from the microcontroller regulate motor speed, allowing smooth acceleration and turning. This module provides the mechanical drive necessary for both movement and cleaning actions.

F. Power Management Module

The power management module is based on a 12 V lithium-ion battery pack with an integrated Battery Management System (BMS). The BMS ensures safe charging, discharging, and overcurrent protection. Voltage regulation is achieved through buck converters, providing 12 V for motors and 5 V for the control and sensor circuits. This efficient power distribution extends operational time while maintaining consistent performance. The rechargeable design supports portability, making the system practical for household and small-scale commercial use.

G. Dual-Surface Orientation Module

The dual-surface orientation module allows FlexiClean to switch seamlessly between floor and wall cleaning modes. This is achieved through suction or magnetic adhesion mechanisms that hold the robot securely against vertical surfaces. When operating on walls, the motors adjust torque to maintain grip and movement stability. This module is central to FlexiClean's innovation, enabling versatile operation across multiple surface types. It ensures balanced weight distribution, prevents slippage, and enhances cleaning accessibility in hard-to-reach areas.

Each of these modules works in synchronization under the control of the Arduino Nano microcontroller, ensuring reliable communication between sensors, actuators, and the mobile app. The modular approach not only improves system scalability and maintainability but also allows future enhancements such as AI-based path optimization and IoT connectivity.

VI. RESULTS AND DISCUSSION

The FlexiClean prototype was successfully designed, assembled, and tested to evaluate its performance in dual-

surface cleaning, mobility, obstacle avoidance, and power efficiency. The experimental setup included both floor and wall cleaning tests on various surfaces such as tiles, glass, and painted walls. The results confirm that the system operates reliably under different environmental conditions with stable Bluetooth connectivity and efficient cleaning coverage.

During testing, the robot was able to navigate and clean a 2 m × 2 m floor area within approximately 7 minutes, consuming minimal power from a 12 V, 2200 mAh lithium-ion battery. The same setup achieved wall cleaning for a height of 1.8 meters with firm surface adhesion using suction-based support. The cleaning efficiency was measured based on surface coverage and dirt removal rate, achieving an average of 92% cleaning effectiveness for dry dust and 85% for liquid stains.

A set of three ultrasonic sensors ensured effective obstacle detection and collision avoidance. The robot successfully identified obstacles at a range of 10–200 cm, dynamically adjusting its motion without human intervention. The Bluetooth control system provided smooth directional movement, and command delay was measured to be less than 200 milliseconds, ensuring real-time response. This responsiveness was validated using the MIT App Inventor-based mobile interface, which allowed easy switching between floor and wall modes.

Table I: Key performance parameters observed during experimental testing.

Parameter	Measured Value	Remarks
Floor Cleaning Area	4 m²	Single charge operation
Wall Cleaning Height	1.8 m	Stable suction
Average Cleaning Efficiency	88–92%	Based on stain type
Battery Backup Time	45–55 minutes	Continuous operation
Obstacle Detection Range	10–200 cm	Accurate sensing
Control Delay (Bluetooth)	< 200 ms	Smooth response

The graphical analysis of cleaning area versus time indicates that FlexiClean maintains consistent performance over prolonged operation, with only a 10% drop in speed after 45 minutes of battery usage. The optimized water and detergent dispensing mechanism reduced fluid consumption by approximately 35% compared to manual cleaning, proving its eco-friendly design.

However, certain limitations were observed. The wall adhesion system showed minor instability on rough, uneven surfaces. Additionally, the robot's suction fan increased power consumption during vertical operation, slightly reducing battery life. These challenges can be addressed through stronger adhesion mechanisms and optimized power distribution circuits in future versions.

Overall, the FlexiClean system achieved its design goals of dual-surface cleaning, mobile control, and cost-effective operation. The results validate that the robot is capable of performing efficient, sustainable, and autonomous cleaning suitable for both household and small commercial environments.

VII. CONCLUSION

The development of FlexiClean demonstrates an efficient approach to modernizing cleaning through robotics. By integrating mobile control, obstacle detection, and dual-surface cleaning, the system reduces manual effort and improves coverage. The prototype validates the feasibility of using low-cost components to build a functional smart cleaning robot for homes and small businesses.

VIII. FUTURE ENHANCEMENTS

The system can be further improved by adding voice control, allowing users to operate it through simple speech commands for greater convenience. Expanding its cleaning capability to work on surfaces such as glass, stone, and metal would make it more versatile. Incorporating AI-based dirt detection using cameras or infrared sensors would help the system identify and focus on dirty areas more effectively. A self-charging dock could also be introduced so the device can automatically return for charging when needed. In addition, integrating IoT and cloud features would enable Wi-Fi control, remote monitoring, and easy connection with smart home systems.

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